Topic: Instruction Formats

Q. What is an Instruction Format? Explain the different types of instruction formats based on the number of addresses (Zero, One, Two, and Three-Address instructions) with examples.

> **Definition to Remember**

> An *
perforr
zero, c

Instruction Formats & Addressing

Used in **Stack-based architectures**. The operands are implicitly located at the top of the stack.

- **F

2. One-Address Instruction

Used in **Accumulator-based architectures**. One operand is specified; the second operand and the destination are implicitly the Accumulator register.

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3. Two-Address Instruction

Used in **General Purpose Register architectures** (like Intel x86). It specifies two addresses, where one usually acts as both a source and the destination.

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4. Three-Address Instruction

Used in modern **RISC architectures** (like ARM, MIPS). It specifies three distinct locations (one destination, two sources). Makes expressions short but requires longer instruction strings.

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> **Must-Write Points (for 10 marks)**

> 1. **

> **Quick Recall**

> `Inst
R2, R3